Visual Working Memory and Change Detection: Reassessing Whole Display Interference.

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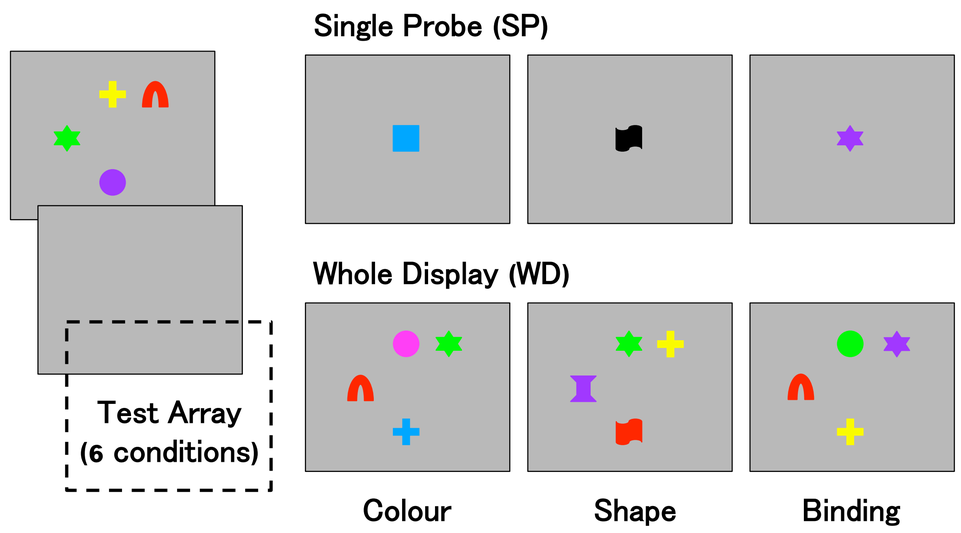
Introduction

Wheeler and Treisman (2002) assessed feature binding in visual working memory (VWM) with two change detection tasks:

- ***Single Probe (SP):*** proportion correct in the binding condition ≈ shape only.
- ***Whole Display (WD):*** binding performance worse than shape only.

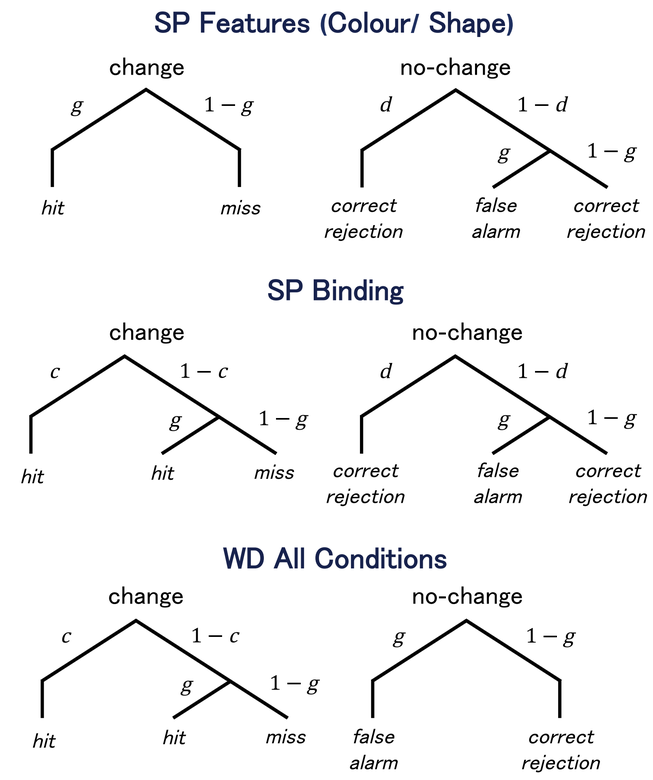
Interpreted as *binding specific WD interference* due to processing demands at test.

However, proportion correct in SP task *overestimates* binding performance (Cowan et al., 2013). This is not the case for the WD.



Tasks used by Wheeler & Treisman (above). Figure shows change trials only, 50% of trials contained no-change (although items shifted locations).

Task processing tree models (right) derived from slots model of VWM (see Cowan et al. 2013). d = probability that probe item is in VWM, c = probability one or both changed items are in VWM, g = probability of guessing 'change'.



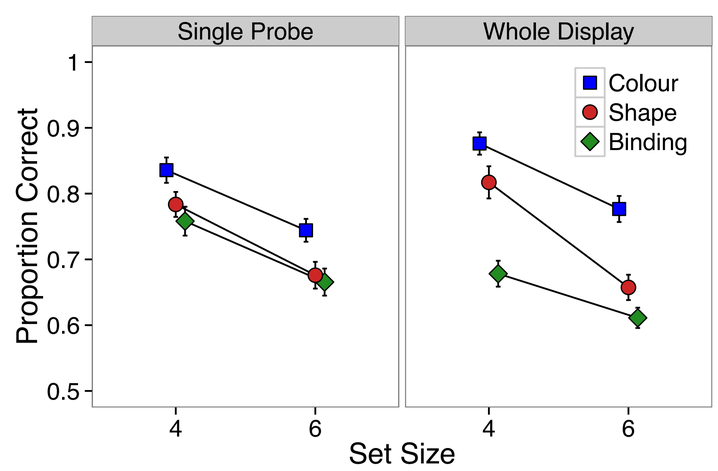
Experiment 1

Compared SP and WD tasks for colour, shape, and colour-shape binding.

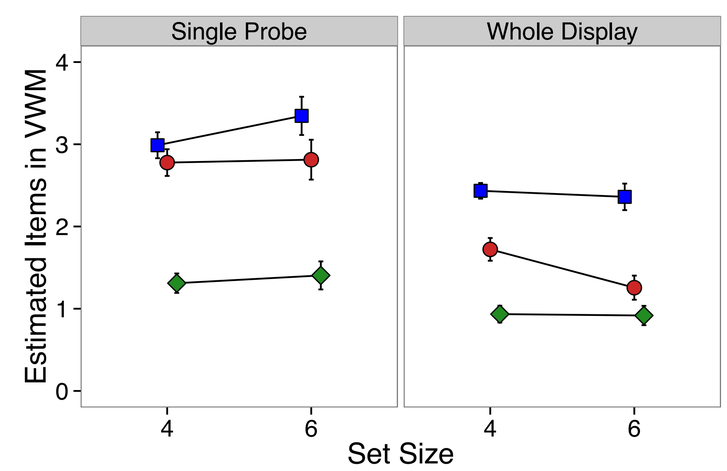
Replicated Wheeler & Treisman's original pattern of proportion correct.

We used processing models derived from the slots conception of VWM (see top right) to better compare tasks.

All experiments N = 24, 72 trials per condition.



⊙No overall difference between SP and WD, 0.000 95% HDI [-0.094, 0.092].§ ⊙Correct response more likely for features vs binding, 0.511 [0.416, 0.603]. ⊙Feature vs binding contrast was larger for WD task, 0.485 [0.297, 0.670] (Wheeler & Treisman, 2002).



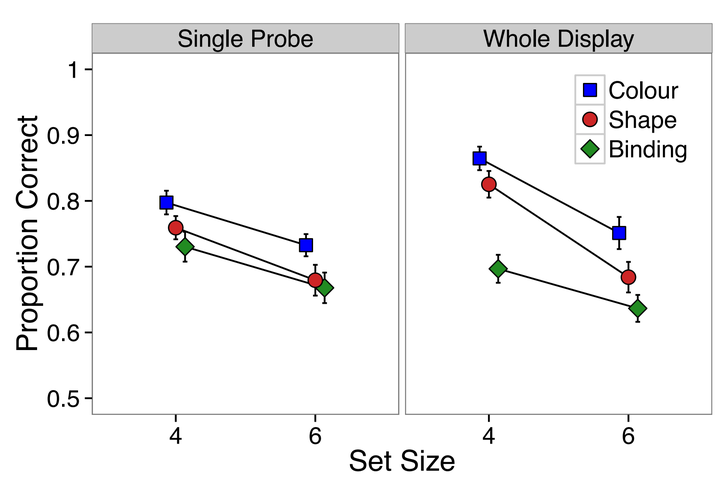
⊙Overall estimates higher for the SP task, 0.825 [0.674, 0.986]. ⊙Estimates were larger for features than bindings, 1.307 [1.141, 1.468] (see also, Cowan et al., 2013). ⊙Feature vs binding discrepancy larger for SP task, 0.552 [0.233, 0.872].

Experiment 2

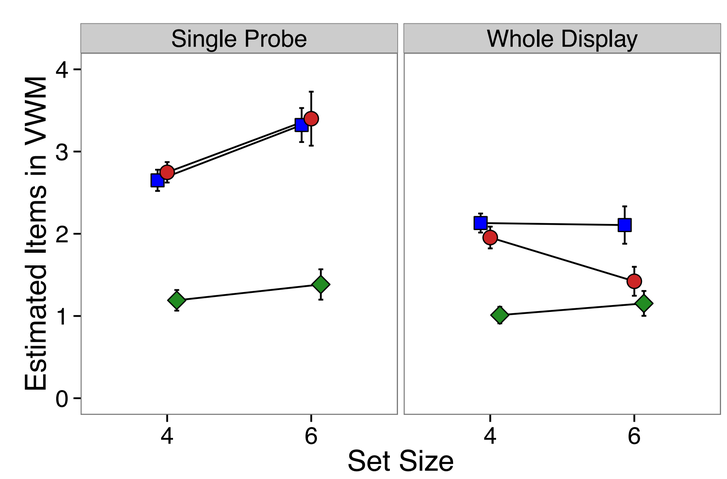
In Exp 1 irrelevant features were presented at test in the colour/ shape only conditions.

Perhaps this exaggerated the difference between features and binding?

Exp 2 removed irrelevant features in colour/ shape only conditions. At test colours presented in squares and shapes presented in black.



⊙Overall lower accuracy with a SP, -0.119 [-0.211, -0.030]. ⊙Again, colour/shape accuracy > binding, 0.431 [0.337, 0.523]. ⊙Difference between feature and binding performance larger for WD task, 0.411 [0.222, 0.597].



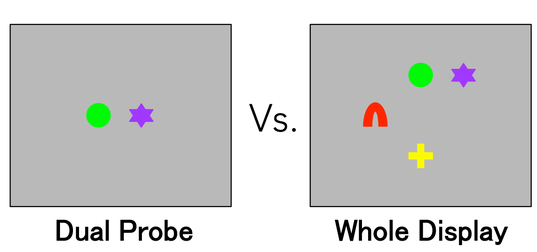
⊙Estimates were larger for the SP task, 0.805 [0.629, 0.983]. ⊙Features > binding, 1.263 [1.073, 1.448]. ⊙Difference between feature and binding larger in the SP task, 0.851 [0.483, 1.221].

Experiment 3

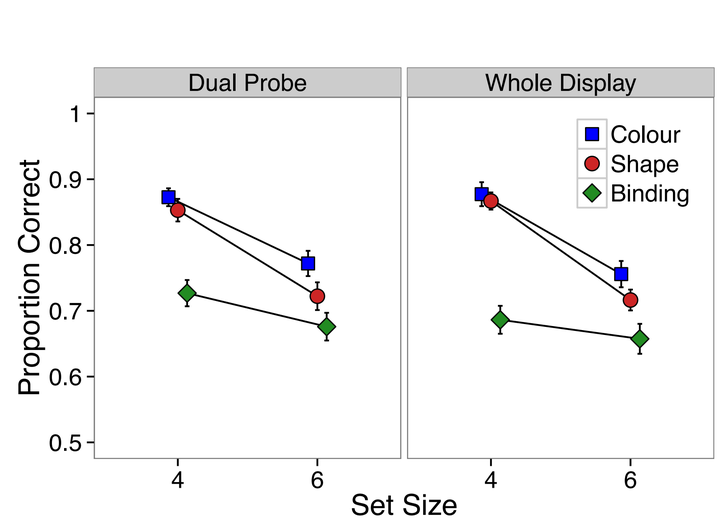
Estimates of the number of items in VWM suggest overall WD interference and larger binding cost in the SP task.

Possible these findings are artefacts from applying different models.

Exp 3 attempted to match the two tasks *directly*.



Replaced SP with 'dual probe' where *both* probe items are the same or *both* are different.



⊙No difference between SP and WD, 0.040 [-0.056, 0.137]. ⊙Features more accurate than binding, 0.692 [0.595, 0.788]. ⊙No clear interaction between probe type and feature vs binding contrast, 0.152 [-0.037, 0.343].

Conclusions

The WD task does not cause binding specific interference, rather the SP task overestimates binding accuracy relative to colour /shape.

Exps 1 & 2 estimated the number of items in VWM using the slots model of VWM. Estimates were lower in the WD task and a disproportionate binding cost was found in the SP task. However, it is possible that this is an artefact of our modelling approach.

Exp 3 attempted to match the tasks directly and found equivalent patterns of performance when a single decision was made at test relative to multiple decisions. Thus any interference caused by multiple test items is minimal and not specific to VWM for feature bindings.

§ Accuracy was analysed with Bayesian hierarchical logistic regression (Kruschke, 2015) estimated with *JAGS* (Plummer, 2013). Contrasts are on log odds scale. Hits and false-alarms were used to estimate number of items in VWM using formulae derived from models described above – analysed using the default priors of Rouder et al. (2012) implemented in the *BayesFactor R* package (Morey & Rouder, 2015; R core team, 2015). Posterior means on specific contrasts are presented with their 95% Highest Density Intervals (HDIs).

References ■Cowan, N., Blume, C. L., & Saults, J. S. (2013). *Journal of Experimental Psychology: Learning, Memory, and Cognition, 39*, 731 – 747. ■Kruschke, J. K. (2015). *Doing Bayesian Data Analysis.* Elsevier, London. ■Plummer, M. (2013). *JAGS version 3.4.0 user manual.* ■Morey, R. D., & Rouder, J. N. (2015). BayesFactor: *http://CRAN.R-project.org/package=BayesFactor* ■R Core Team. (2015). *http://www.R-project.org/* ■Rouder, J. N., Morey, R. D., Speckman, P. L., & Province, J. M. (2012). *Journal of Mathematical Psychology, 56*, 356–374. ■Wheeler, M. E., & Treisman, A. M. (2002). *Journal of Experimental Psychology: General, 131*, 48 – 64.

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